GSOC Proposal: Inkscape, 2023

Improving CSS Stylesheet Support

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About me

Personal Information

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Education Information

- University: Indian Institute of Technology, Roorkee
- Major : Mathematics and Computing
- Current Year: Sophomore (2nd year)
- Degree : Bachelor of Science (BS)

Background

I am Vansh Uppal, UG Sophomore majoring in Mathematics and Computing at Indian Institute of Technology, Roorkee. I have been involved in various software development and tech-related endeavors in the form of hackathons, CTFs, course projects, and projects at SDSLabs (a student technical group at IITR I am a part of)

As a member of SDSLabs, I have learnt a lot and have gained valuable experience as a developer and have worked on various projects on a variety of programming languages

Programming languages

- **Languages:** C, C++, Python, Bash, HTML, CSS, Javascript, Golang, PHP, FORTRAN
- **Software Packages:** Git, Docker, Kubernetes, Figma, Vim, MySQL, Visual Studio, Linux, CLion
- **Others:** Object-Oriented Programming, Analysis and Design of Algorithms, Statistical Machine Learning, Numerical Analysis, Discrete Mathematics

Why me?

I have a lot of experience contributing to open source projects and strong design experience and have been coding in Python, C and C++ for a long time. Moreover I have experience working in SDSLabs where we build production-level applications from scratch and thus I know the technical difficulties that might arise and how to deal with them.

Achievements

- **Syntax Error, 2022**: 2nd Rank in Overall Category
- **D3Code, 2022:** Honorable Mention
- **Mathemania, 2023:** Honorable Mention

Past Projects

Katana

- Katana is one of its kind Attack and Defence CTF hosting platforms currently being developed by SDSLabs. The project aims to make hosting an attack and defense CTF easy and uses concepts of Kubernetes, Gogs , Golang, Python and more. Github
- My contribution to this project has been: Working on flask applications, fixing bugs, Working on kubernetes and docker internals and more to make the hosting experience as smooth as possible PR:46 PR:5
- I am also working on it's frontend which is being written in Nuxt.js (Github)

Rootex

- Rootex is an advanced C++ 3D game engine powering an in-production game yet to be announced. The game will finally ship on Windows and use DirectX 11. The engine is based on the Entity-Component-System architecture and uses a Lua scripting Engine for making games. GitHub | Docs
- My contribution to Rootex covers creating the in-development game to be shipped with Rootex, fixing bugs in the engine, and tweaking it for the game.
- The engine was also covered by GamesFromScratch (Link)

Air-Guitar

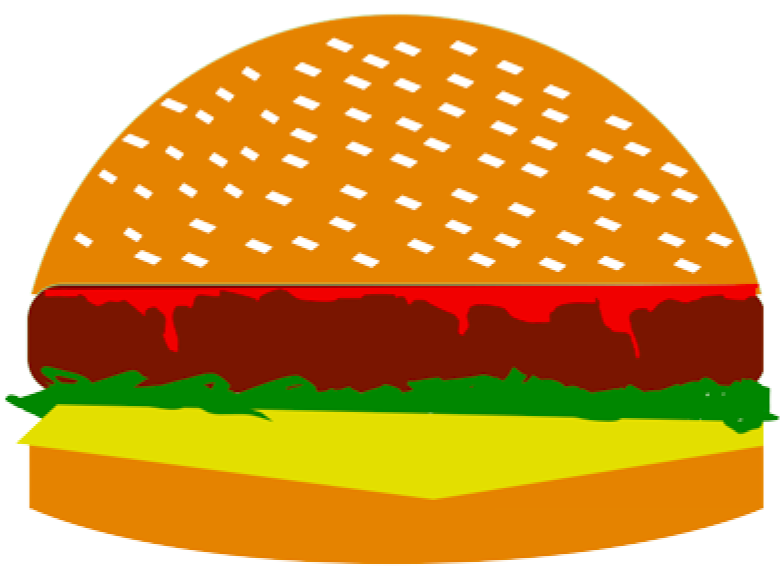
- A python project I made as part of a team during Syntax Error (Github) Hackathon. This project utilized open-CV to capture video in real time
- The project utilized a media-pipe library to detect hand gestures and positions.
- This project earned me 2nd rank in Syntax Error, Overall Category

Journey with Inkscape:

I had heard of and used inkscape since school since for various development projects and websites, I usually had to create SVGs on my own and this was an easy to use and free software. However it was only in late January / Early February that someone pointed out to me that inkscape is in fact an open source organization and I can contribute to it. Having experience in C++, Python and knowing the basics of Computer Graphics, I started contributing and have been active in the community ever since.
One of the things I like most about the Inkscape community is how helpful the people are. They have helped me become a better developer, from code quality to improving my commit messages, to in general giving me ideas on how to become a better developer.

I feel I am suitable for this project because of my experience working in a development community (SDSLabs) for the past year. I have good knowledge of C++ and python and having worked on projects based on it, understand the technical difficulties that come with it and how to tackle various problems that arise.

A burger when I first tried my hands on inkscape (pardon my artistic skills)



Contributions:

I started contributing on inkscape and learned about easy fixes and tried my hands on those. It was fun working on various issues and solving them and an exhilarating feeling seeing something that I wrote become part of an app that people use on a daily basis.

Then I started taking up better issues and trying my hands on them.

- Fix gaps calculation on duplicated objects in LPE Tiling**(**Status: **Merged) [#5040]**
- Fix Paste Style being applied on locked layers too**(**Status: **Merged) [#5044]**
- Fix indentation **(**Status: **Merged) [#5062]**
- Increase max_pixel_size to 80. This allows for up to 500% icon scaling**. (**Status: **Approved) [#5134] (**Issue resolved as asked, however was pointed out to maybe increase the scope and deal with a bigger problem associated with this)
- Unset color set my attribute **(**Status: **Open) [#5136] (**Currently waiting for further reviews)

Project Details

Project Idea

Improve the CSS Stylesheet support for inkscape. This involves UX improvements as well as feature improvements. This is so that users that use the CSS Stylesheet feature find it more intuitive and better to work with.

Initial Research

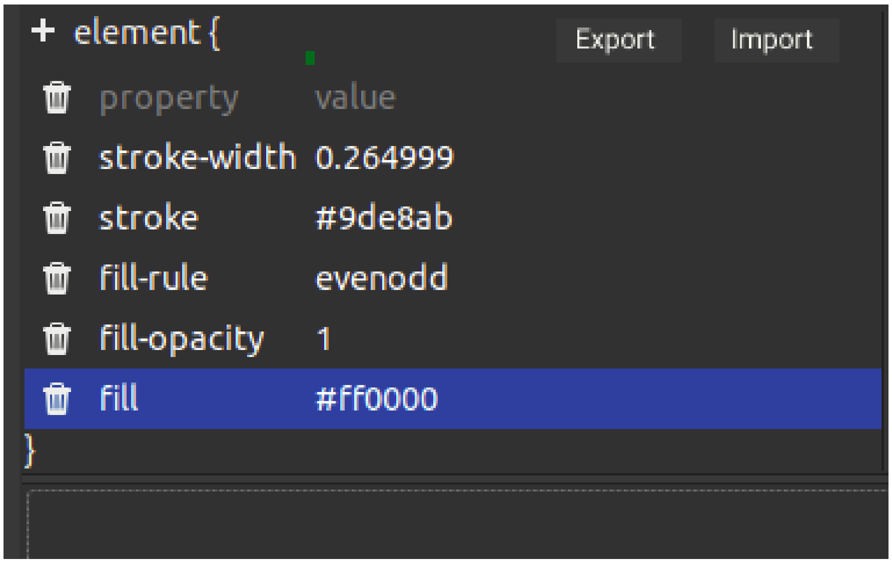
When I first saw this project idea on the GSoC website, I immediately thought of exploring new and similar ideas that existed in other softwares.
The closest and best match I thought of relating this to was Blender since it has a somewhat similar concept of objects and layers.
I also incorporated ideas for different sources like VSCode SVG maker (https://marketplace.visualstudio.com/items?itemName=jock.svg) and drew experiences from various different tools I have used.

I took the help and guidance of 2-3 UX designers of SDSLabs and also had around 4-5 of peers run inkscape and utilize the CSS feature so I can have their inputs too.

After all of this, I came up with a total of 4 ideas (Originally 6 ideas which I can add as stretch goals) and concepts that I can implement which will enhance the UX improvement phenomenally.

Proposed Features and Implementation

- **Import/Export:** Allow users to create a CSS style sheet as an external file and allow them to import and apply it to other projects as well. This would help in creating reusable CSS stylesheets. [Using random font due to lack of inkscape standard font for figma mockup]



Implementation

The style properties are declared in this file.

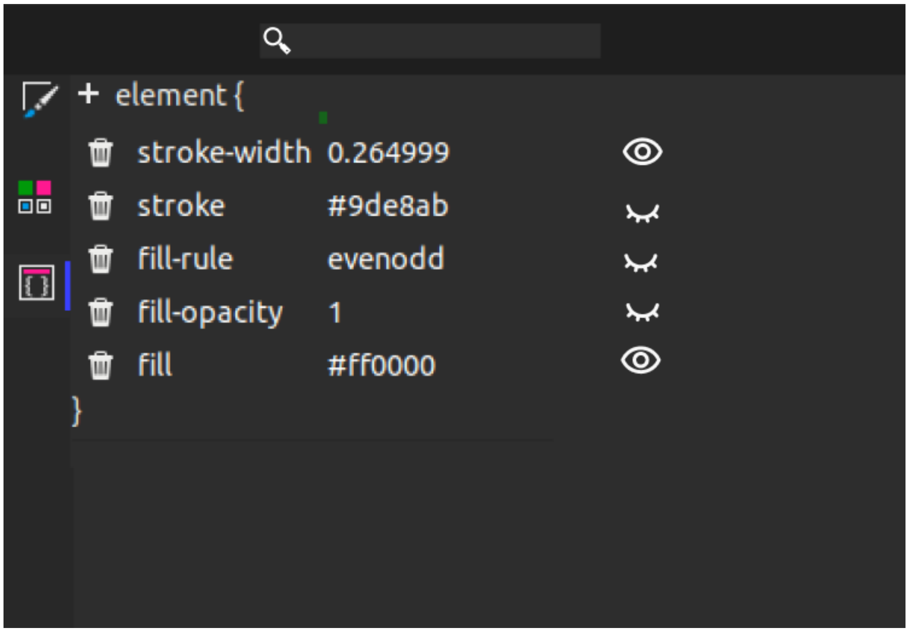
Upon export, I'll create a new file, asking for the name of the CSS file the user wants to set and the iterate through the properties to create an accurate CSS file

On import, I'll be traversing the CSS file line by line and adding the properties.

Should I overwrite the pre-existing CSS while importing or appending is something that I intend to discuss with my inkscape mentors.

- **Ignore CSS:** Users can selectively unset a particular attribute of an object and see the change it would have. Combined with Nested CSS (see next point), this would considerably help the users.

  This is inspired from a similar feature in the inspect element of Chromium based web browsers.



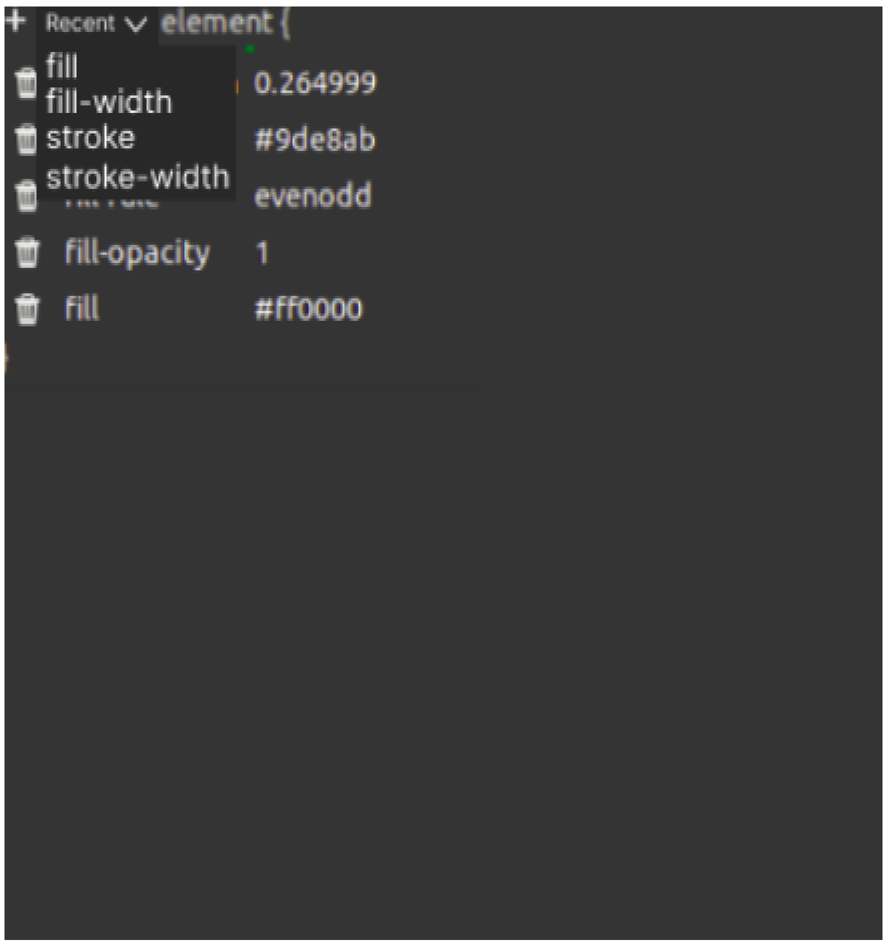
Implementation: The above feature will be easily implemented. Having already worked on the Attribute unset feature, This will be a simple toggle.

```
if(unset){
  setAttribute(attribute_name,nullptr)
}
else{
  setAttribute(attribute_name,attribute_value)
}
```

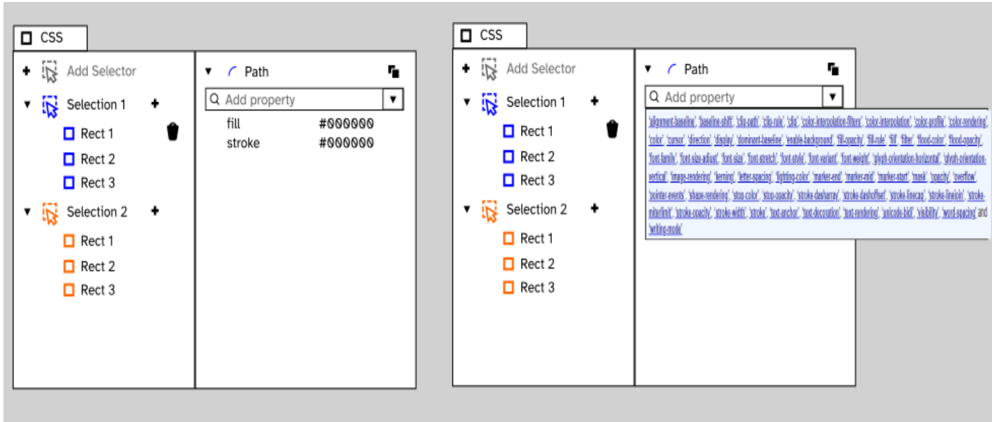
Having made mockups in figma, I'll export and use the icons myself or if the designers make one, will integrate the icon using GTK.

Another approach is to replace the functionality with one that comments out that particular CSS. Exact implementation can be discussed later.

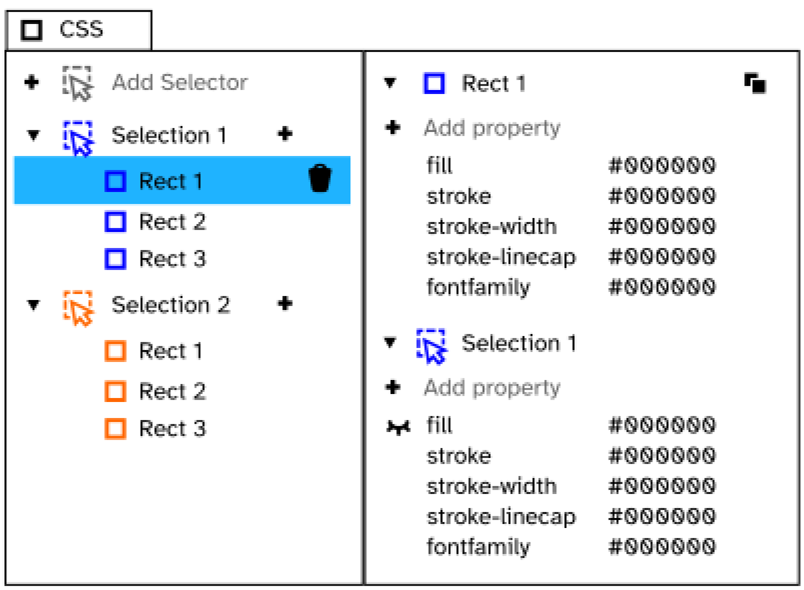
- **Recent Attributes:** Including Recent tab at the top of Selectors tab allow users to access properties they most recently used so that instead of having to write fill for each object, they can just select it from the recents tab.



This was inspired by Adam.Bellis' idea for a drop down to add properties. I personally feel that would create too big a list for it to be comfortable for the users to use. However more user research is needed about which direction to go in when it comes to this



- **Nested CSS:** Sometimes while using nested objects (see image below), it's necessary for the users to see what properties they inherit from the parent and thus see the impact of the same on the children. So we can display both those CSS making it more convenient to be used by the user (Similar to what is used for web pages CSS)



Implementation: This will be a simple recursive algorithm which will display the parent css. This will be a small change in this function.

I will iterate through the parents and display the attributes.

```
for (auto child : children){
  auto par = child->parent()
    if (par->getRepr()->attribute("style")) {
        Glib::ustring style = obj->getRepr()->attribute("style");
        attr_prop = parseStyle(style);
        for (auto iter : par->style->properties()) {
          if (attr_prop.count(iter->name())) {
              auto value = attr_prop[iter->name()];
              empty = false;
      ...
  }
  }
```

A rough code example of how it will be implemented

Use Cases:

- Increase user experience while using CSS Stylesheet by giving them access to recent attributes so they dont have to manually add each attribute every time for a new object if they frequently use some attributes
- Giving users access to ignore CSS will be especially helpful as now to check impact of a particular attribute, you have to delete the entire attribute and rewrite. This is exhausting sometimes hence this would be very helpful among users.
- Giving users the option to export and import CSS file on certain elements allows reusability of stylesheet and thus would help more users to utilize this feature for multiple projects instead of rewriting CSS from scratch.
- Other features just basically increase UX and avoid confusion among users (especially naive users) when using the CSS Selectors option, like the nested CSS option

Timeline

I plan to do a long project under inkscape. To have time available for the same, I had all my pending projects shifted from summer to April giving me ample time to contribute

I have decided on the following timeline

Date (from and to)	**What I plan to do**
Pre GSOC Period	
4th April to 4th May	Solve issues and help inkscape be ready for 1.3v release. There are about 32 issues that haven't been started yet so work on them.
GSOC Period	
4th May to 28th May, 2023 [**Community Bonding**]	Discussion with mentors about the project and discussion about the first milestone. Talk to other developers and designers to know if they have some vision for the CSS Stylesheet too which I can incorporate. Learn more about inkscape code base and the relevant areas I'll be changing
Coding Period Begins	
28th May to 12th June	Implement **recently used CSS features**. For first time users it will show the popular choices and then change according to whatever CSS attributes they do use. I'll keep taking feedback from developers and designers about the changes and how they can be made better

13th June to 30th June	**Implement Import and export CSS** feature.
1st July to 4th July	Documentation of work done till now and buffer days (in case of any issue taking more time than expected)
Phase 1 Eval and Report (4th to 7th July)	
7th July to 21st July	**Implement nested CSS feature**
21st July to 1st August	**Implement the ignore CSS feature**
1st August to 3th August	This is when I provide the inkscape developers the first version of all the changes I have done so that they can try out the new UX and give feedback or recommendations (if any)
4th August to 21st August	Implement any additional feature request or changes that I might receive as feedback.
21st August to 28th August	Documentation and Final Evaluation and submitting my report.
Stretch Goal	Add support for visibility attribute in inkscape

Post GSoC Goals

Contributing to Open Source organizations was on my to-do list for a long time. I think I have gotten over the initial learning curve of getting familiar with the code base at Inkscape now that I am pretty comfortable with the coding practices here at Inkscape, I plan on continuing to contribute to Inkscape even after GSoC along with solving any other issues or maintenance needs that come up in the future regarding the above mentioned features.Also, I also aim to add support for visibility attribute to inkscape.

Project Summary

My project (Improving CSS Stylesheet Support) implements 4 functionalities which would improve UX significantly: **Nested CSS**, **the Ignore CSS, Import and export CSS** and **Recently used Attributes.**

These features are necessary for inkscape to create a more user friendly environment.

Some have been drawn from what Adam.Bellis recommended and some are based on my personal views as well as experience of some of my peers who use inkscape.

Apart from this, I am also open to suggestions of other developers and designers as to how I can make this project better.

Availability

II can dedicate more than 30 hours a week. The college's summer vacations are scheduled from mid-May to July. Therefore, there would be no classes for the first two months of the program's coding period and no exams throughout the coding period. My college will reopen on July 18th. During this time, I could work more than 20 hours a week.

I generally work from 4 PM to 5 AM IST (6:30 AM to 7:30 PM Eastern Time), although I find it flexible to adjust my working hours if required.

Other than this project, I have no commitments/vacations planned for the summer. I shall keep my status posted to all the community members every week and maintain transparency in the project. [To accommodate for the same, I have already shifted and completed all my projects in the month of March and April]

I am 100% dedicated to inkscape and have no plans whatsoever to submit a proposal to any other organization.